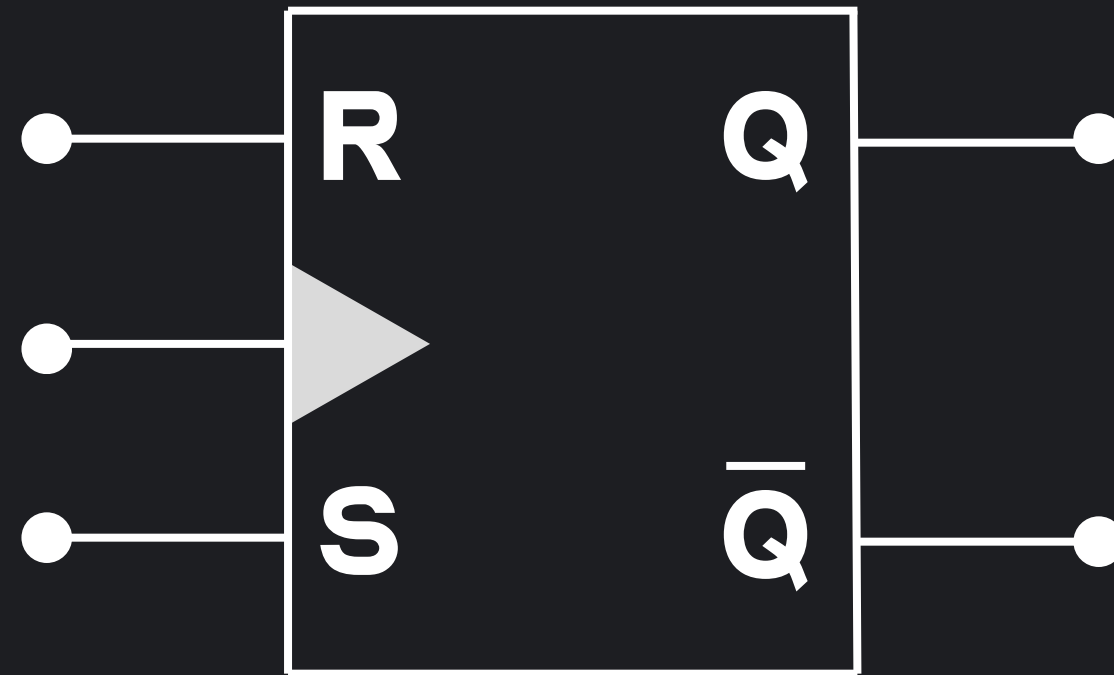


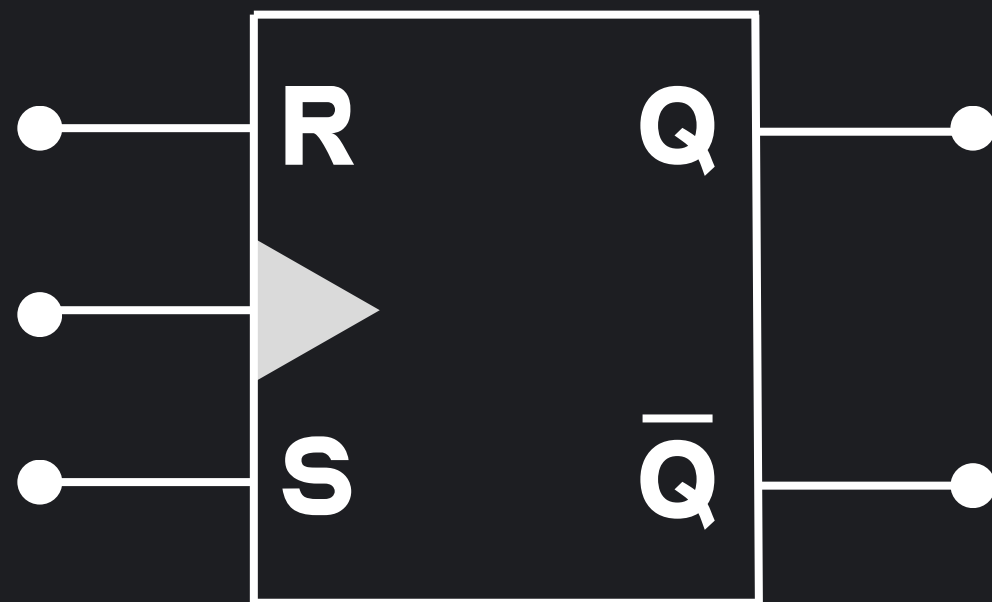
RS-FLIP FLOP



What is a RS-FLIP FLOP?

An RS Flip-Flop is a basic sequential logic gate that produces an output based on its inputs and its previous state. The output is in a Set state when the Set input (S) is HIGH (1), while it maintains its previous state when both inputs (R and S) are LOW (0).

Truth Table



S	R	CK	Q	made of operation
x	x	↓	Q_0	NO Change
0	0	x	Q_0	NO Change
0	1	↑	0	Resets
1	0	↑	1	Sets
1	1	↑	Q_0	Change

Function of the RS-FLIP FLOP

An RS Flip-Flop performs a logical storage operation on its inputs. The main function of an RS Flip-Flop is to serve as a basic memory element and produce an output signal according to this rule:

Output = No Change when both inputs R and S = 0.

Output = 1 (Set) when S = 1 and R = 0.

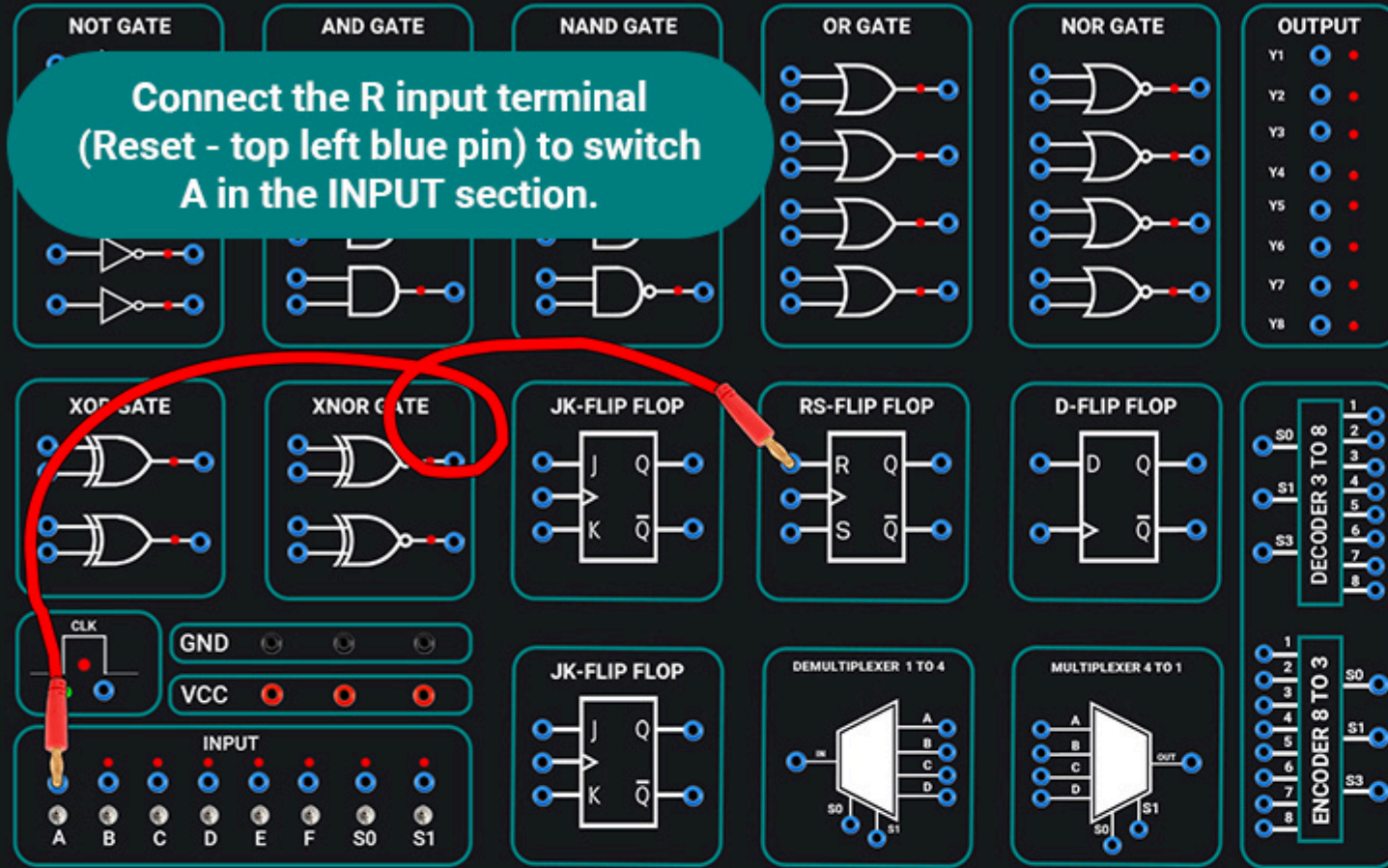
Output = 0 (Reset) when R = 1 and S = 0.

Output = Invalid/Not allowed when both inputs R and S = 1.

1

SMART DIGITAL CIRCUITS BOARD

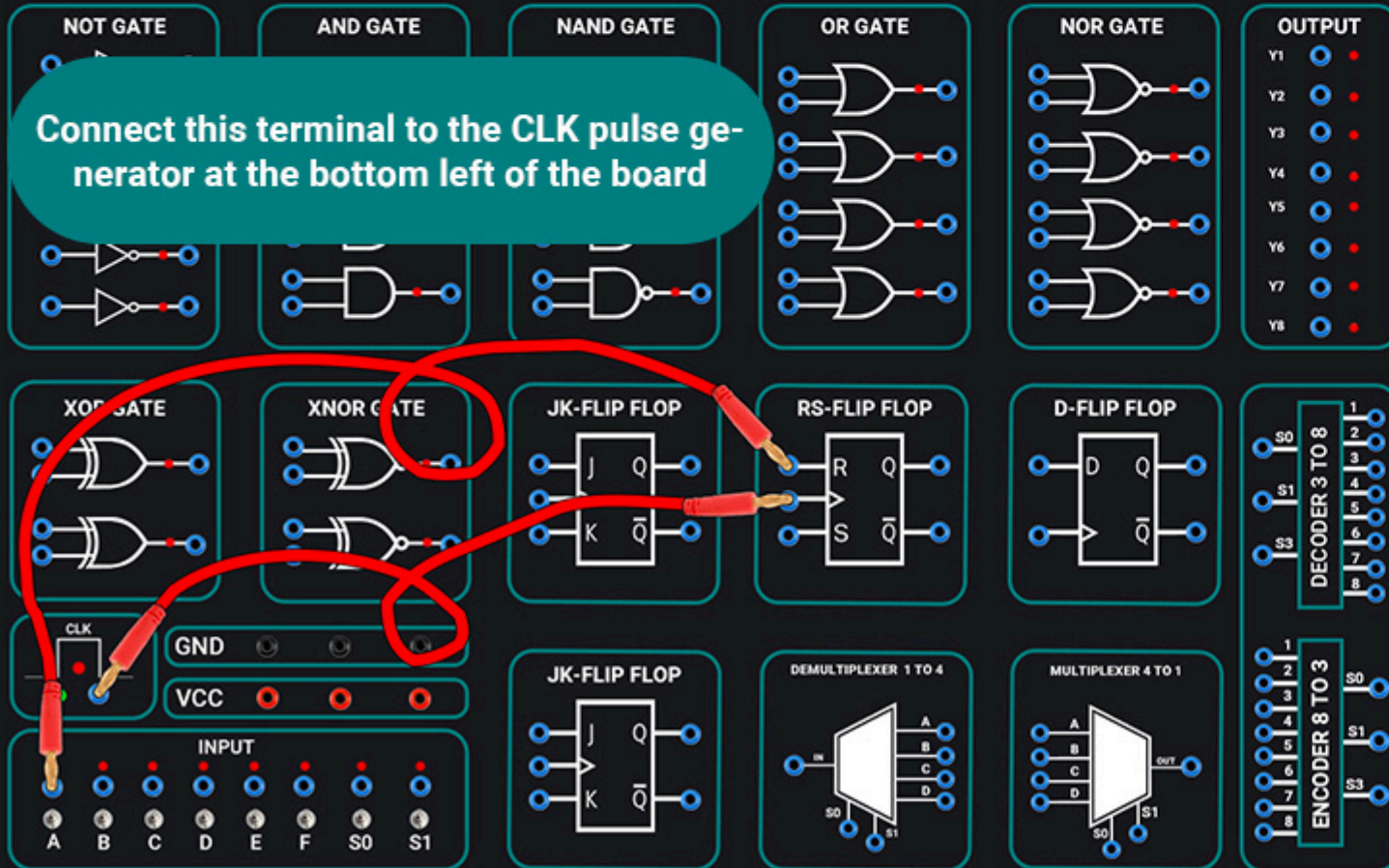
Connect the R input terminal (Reset - top left blue pin) to switch A in the INPUT section.



2

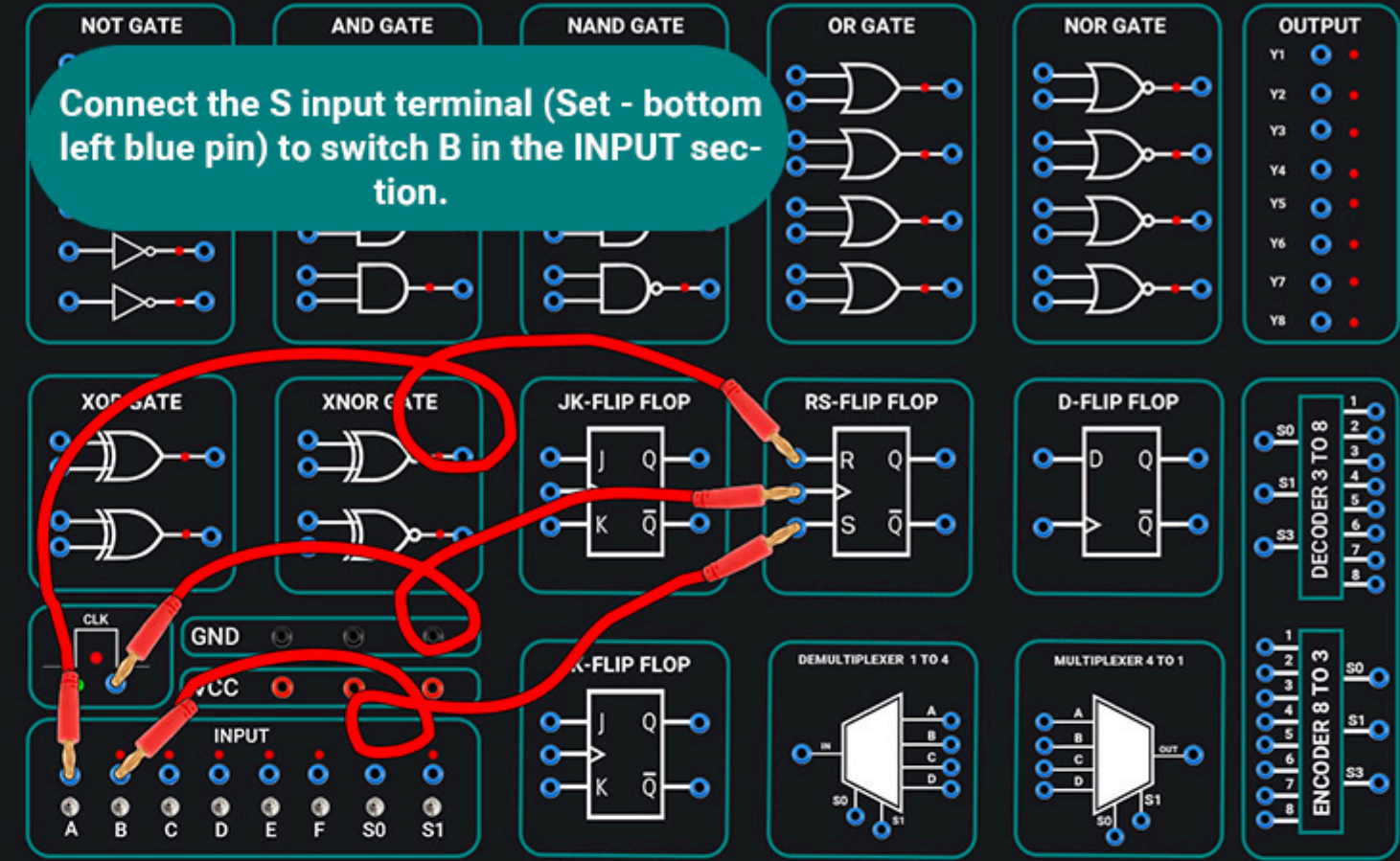
SMART DIGITAL CIRCUITS BOARD

Connect this terminal to the CLK pulse generator at the bottom left of the board



SMART DIGITAL CIRCUITS BOARD

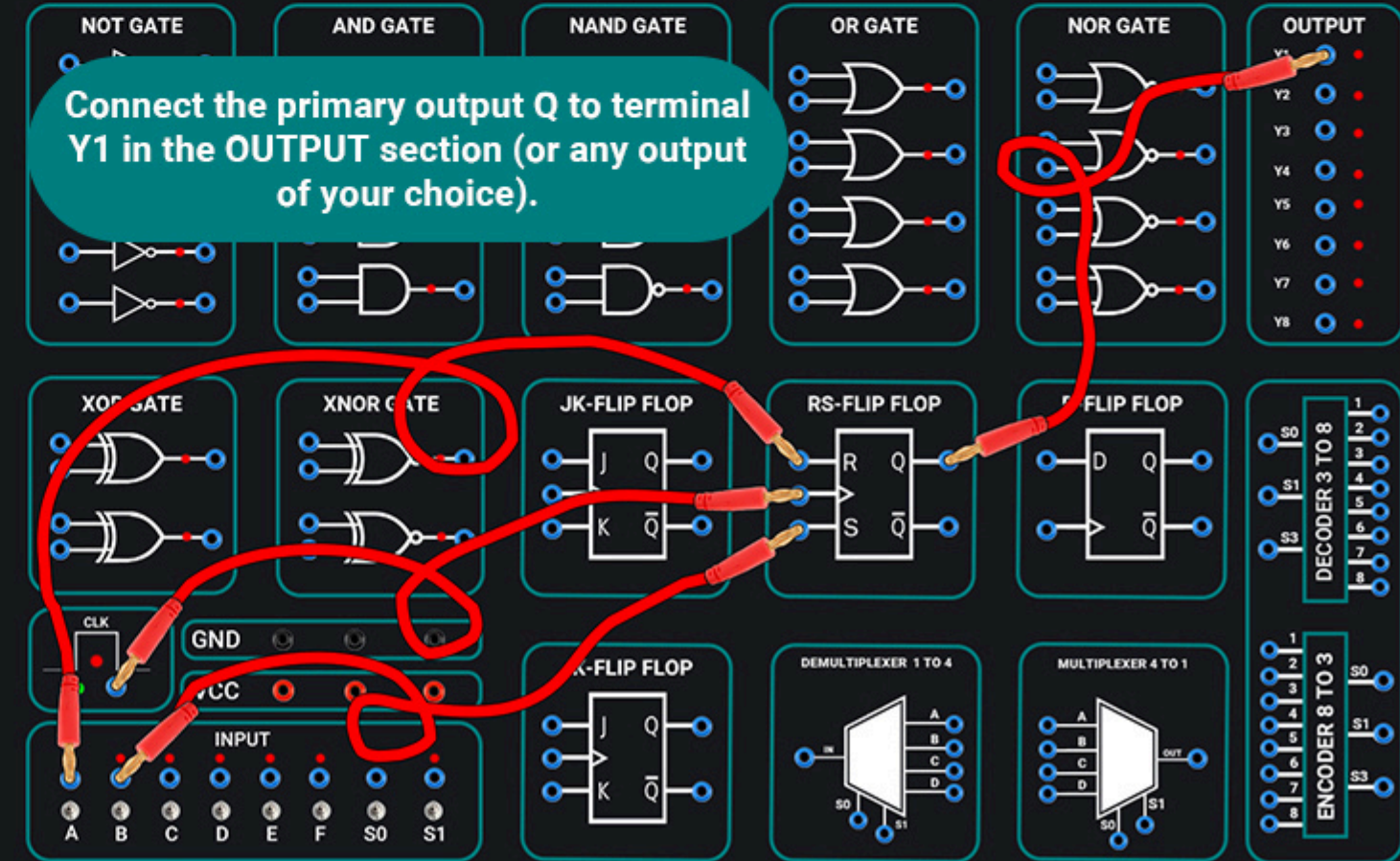
Connect the S input terminal (Set - bottom left blue pin) to switch B in the INPUT section.



3

SMART DIGITAL CIRCUITS BOARD

Connect the primary output Q to terminal Y1 in the OUTPUT section (or any output of your choice).



4

SMART DIGITAL CIRCUITS BOARD

Connect the complementary output $\bar{Q}$ to terminal Y2.

5

